An Integrated LSTM-TabNet Ensemble with Elastic Net for Jakarta Composite Index Forecasting

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***Abstract*—Forecasting the Jakarta Composite Index (IHSG) remains challenging due to its high sensitivity to global financial dynamics and nonlinear market interactions. This study proposes an integrated ensemble framework combining a Long Short-Term Memory (LSTM) network and a TabNet model, with an Elastic Net meta-learner for final prediction aggregation. The LSTM captures temporal dependencies in sequential IHSG data, while TabNet processes engineered tabular features to enhance interpretability. Hyperparameters for all models are optimized using the Optuna framework. The model is trained on daily data from January 2010 to January 2025, incorporating global stock indices, commodity prices, exchange rates, and macroeconomic indicators. Experimental results on an unseen test set show that the proposed LSTM-TabNet-ElasticNet ensemble achieves a Root Mean Squared Error (RMSE) of 49.89 and a Symmetric Mean Absolute Percentage Error (SMAPE) of 0.53%, outperforming standalone LSTM (RMSE 50.29), TabNet (RMSE 50.03), and serial TabNet-LSTM hybrids (RMSE 63.59). Feature importance analysis highlights lagged returns (S&P 500, Gold) and volatility (VIX) as dominant predictors, emphasizing the influence of recent global market movements on IHSG dynamics. The proposed framework offers a robust and interpretable approach for financial time-series forecasting in emerging markets.**

***Index Terms*—IHSG, Long Short-Term Memory, TabNet, Hybrid Ensemble, Elastic Net, Financial Time Series Forecasting.**

I. Introduction

The Jakarta Composite Index (IHSG) is a vital indicator of Indonesia's economy. In an interconnected global market, the IHSG is increasingly sensitive to international dynamics; fluctuations in the S&P 500, commodities, and macroeconomic shifts introduce complex nonlinear effects [1], [2]. Accurate forecasting is paramount for investors mitigating risk and for policymakers monitoring economic stability.

Traditional econometric models often fail to capture the complex, non-stationary nature of financial markets. To address this, we propose a novel *parallel stacking* framework integrating an LSTM (for raw temporal patterns) and a TabNet model (for engineered tabular features), aggregated by an ElasticNet meta-learner to handle correlated predictions and prevent overfitting [3], [4], [5]. This approach contrasts with common serial hybrids (e.g., ARIMA-LSTM) by avoiding information bottlenecks. Our contributions are threefold: 1) designing a parallel stacking ensemble for IHSG forecasting, 2) demonstrating its superior accuracy over standalone and serial architectures [6], and 3) providing interpretable analysis identifying recent global volatility and lagged returns (S&P 500, Gold) as dominant predictors.

II. Literature Review and Research Gap

A. Current State of Financial Time-Series Forecasting

Financial time-series forecasting has evolved significantly with the adoption of machine learning architectures that address the limitations of traditional econometric models. Recent advances can be categorized into three main approaches: deep learning models for temporal dependencies, attention-based models for tabular data, and hybrid architectures that combine multiple paradigms.

Deep Learning and Attention-Based Models: Deep learning models, such as Long Short-Term Memory (LSTM) networks, have become a cornerstone for capturing temporal dependencies in financial data [4], [7], addressing issues like vanishing gradients and enabling the capture of long-range patterns. However, these models face challenges with high-dimensional feature spaces, limiting their standalone effectiveness. To address this, attention-based architectures like TabNet [8], [9] have been introduced for tabular data, using sequential attention mechanisms to select features at each decision step. This mimics decision tree logic while maintaining end-to-end differentiability. TabNet excels at feature selection and interpretability, making it highly effective in stock market forecasting, where diverse economic indicators must be integrated.

Hybrid and Ensemble Approaches: Recognizing that individual architectures capture different aspects of financial dynamics, researchers have explored hybrid models. He et al. [3] demonstrated that ARIMA-CNN-LSTM combinations could leverage both statistical patterns and deep temporal features. Similarly, tree-based ensembles like Random Forest remain robust baselines for feature-based forecasting [10], particularly when temporal ordering is less critical than cross-sectional feature interactions. The challenge lies in effectively combining these complementary strengths without introducing architectural bottlenecks.

B. Macroeconomic Influences on Emerging Market Indices

The Jakarta Composite Index (IHSG), as an indication of emerging markets, demonstrates significant susceptibility to domestic and global economic influences. Keswani et al. [11] established that macroeconomic fundamentals, such as GDP growth and inflation substantially, affect the Indian NIFTY index via cointegration connections. Their findings are pertinent to the IHSG context, as Indonesia possesses analogous traits to India: both are substantial rising economies with indices sensitive to foreign investment movements and commodity price variations. Research focused on the IHSG [1], [2], [12] substantiates significant spillover effects from the S&P 500, commodity markets (oil, gold, palm oil), and the VIX volatility index, thereby affirming the relevance of these variables as predictive factors.

C. The Limitations of Serial Hybrid Architectures

While hybrid models promise to combine complementary strengths, their architectural design critically impacts performance. Wei et al. [13] proposed TabLSTM, a serial integration where TabNet performs feature selection before feeding processed representations to an LSTM. Although this approach demonstrated improvements over standalone LSTMs, the serial design introduces an inherent *information bottleneck*. Specifically, TabNet's attention mechanism, optimized for tabular feature interactions, may inadvertently filter or compress temporal patterns that are essential for the LSTM's sequence modeling. When the LSTM receives TabNet's processed output, it operates on an abstracted representation that has already undergone dimension reduction and attention-based filtering—potentially losing raw temporal dynamics such as intraday momentum shifts or short-term volatility patterns that the LSTM could have captured from the original sequences. This architectural constraint, while not explicitly quantified in existing literature, aligns with information theory principles where sequential processing stages can progressively degrade information content [14].

D. Research Gap and Proposed Contribution

Despite the proliferation of hybrid models in financial forecasting, a critical gap persists: **existing architectures predominantly employ serial integration, which risks degrading the unique information each model type can extract**. While stacking ensembles—where diverse models operate in parallel and their predictions are combined via a meta-learner—have proven effective in other domains [6], their application to stock market forecasting with LSTM-TabNet combinations remains underexplored. Parallel stacking fundamentally differs from serial hybrids by allowing each base model to process data independently, preserving both raw temporal sequences for the LSTM and engineered tabular features for TabNet. An Elastic-Net meta-learner can then optimally weight their predictions, leveraging regularization to handle potential multicollinearity while preventing overfitting [5].

Table I
KEY RESEARCH VARIABLES

Variable Name	Source	Variable Name	Source
IHSG (ĴKSE)	Yahoo	Crude Oil (BZ=F)	Yahoo
S&P 500 (ĜSPC)	Yahoo	Gold (GC=F)	Yahoo
VIX (V̂IX)	Yahoo	Natural Gas (NG=F)	Yahoo
Hang Seng (ĤSI)	Yahoo	Coal (MTF=F)	Yahoo
Nikkei 225 (N̂225)	Yahoo	FTSE 100 (F̂TSE)	Yahoo
Palm Oil (EWM)	Yahoo	USD/IDR (IDR=X)	Yahoo
Fed Rate	FRED	BI Rate	FRED

This research addresses the gap through three contributions: First, we introduce a parallel stacking ensemble specifically designed for IHSG forecasting, where LSTM (operating on raw sequences) and TabNet (operating on engineered features) function as independent base learners. Second, we rigorously compare this architecture against both standalone models and the serial TabLSTM approach [13], providing empirical evidence of parallel stacking's superiority for this forecasting task. Third, we employ Optuna-based hyperparameter optimization [15], [16] across all architectures to ensure fair comparison, and provide interpretability analysis identifying which global economic indicators most strongly drive IHSG movements. This framework offers emerging market practitioners a methodologically sound, interpretable tool for navigating increasingly interconnected financial markets.

III. METHODOLOGY

A. Data Acquisition and Preparation

The dataset comprises daily IHSG prices and global indicators (Jan 2010 - Jan 2025) from Yahoo Finance and FRED (Table I). This 15-year period captures diverse market regimes, including the 2013 'Taper Tantrum' and COVID-19 volatility.

Data cleaning involved handling minimal missing values using forward-filling (`ffill`) to maintain temporal sequence and prevent look-ahead bias. Features were then engineered, including percent changes (returns) and technical indicators (MACD, RSI, SMA). The dataset was split 80/20 into training (3224 samples) and test (807 samples) sets chronologically, preserving temporal dependencies. Features (X) were normalized using `MinMaxScaler` to [0, 1], while the target (y, 1-day return) was normalized using `StandardScaler` (zero mean, unit variance) to stabilize training.

To prepare data for sequential models (LSTM, TCN) and for internal validation, we used two time-series cross-

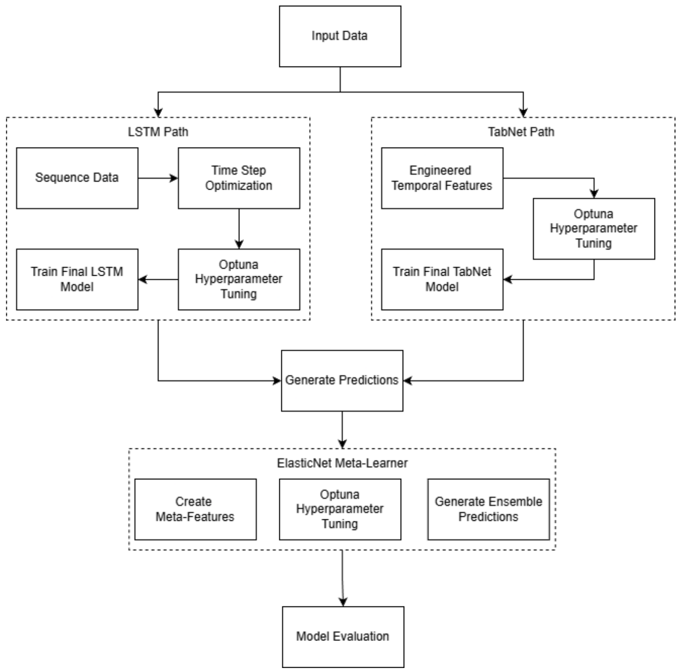

Figure 1. Proposed Ensemble Architecture.

validation techniques. First, for hyperparameter tuning, a `TimeSeriesSplit` strategy with 3-folds was employed on the training set. A 3-fold split was chosen as a balance between obtaining a robust validation score and maintaining a reasonable computational cost [17], [18]. Second, we determined the optimal look-back period using a sliding window. We evaluated window lengths of 30, 60, and 90 days. A 90-timestep window was ultimately selected, as this length (approximately 4.5 trading months) yielded the lowest validation loss during preliminary model testing. Selecting 90 days represents a trade-off: it is long enough to capture significant market cycles (broader patterns) but short enough to avoid overweighting obsolete information (recent dynamics) [19]. This windowing process transformed the 2D data into a 3D sequential dataset [samples, 90, features].

B. Variable Selection Rationale

The variables in Table I were selected based on established financial literature. **Global Indices** (S&P 500, HSI, etc.) represent major economic centers with spillover effects [1], [2]. **Commodities** (Oil, Gold, etc.) directly impact Indonesia's trade balance and corporate earnings, with Gold and Oil also acting as global risk indicators [20], [21], [22], [23]. The **VIX** measures global market fear, which is critical during crises [24], [25]. Finally, **Macroeconomic Indicators** (Fed/BI Rates, USD/IDR) represent monetary policy and currency risk, which are key drivers for foreign investment [11].

C. Constituent Models

We utilize standard LSTM networks [7] to capture long-range temporal dependencies in the raw 90-timestep sequential data. Temporal Convolutional Networks (TCN) are also employed, utilizing dilated causal convolutions to efficiently model long-range patterns with parallelizable computation. TabNet [8], [9] is employed to process engineered tabular features [26], providing inherent interpretability. Predictions from these models are aggregated using an Elastic Net meta-learner [5], which combines L1 and L2 regularization to handle correlated predictions and prevent overfitting.

D. Hyperparameter Optimization

We utilize Optuna, a Bayesian optimization framework [10], [15], [16]. Optuna uses a Tree-structured Parzen Estimator (TPE) to efficiently search the hyperparameter space. The optimization process involved multiple trials (e.g., 20-30 trials per model) searching over a predefined parameter space. This included wide ranges for key hyperparameters such as learning rates (e.g., 10^{-4} to 10^{-2}), batch sizes (e.g., 32 to 512), network depth (e.g., LSTM layers 1-3), and regularization (e.g., dropout 0.1-0.3). This computationally intensive process, requiring significant GPU hours, was essential to ensure that each base model operated at its peak capability before being integrated into the ensemble.

E. Proposed Ensemble Architecture

The proposed framework (Fig. 1) is a two-level stacking ensemble. Level 0 models (LSTM, TabNet) are trained in parallel. Their predictions train the Level 1 Elastic Net meta-learner. Final hyperparameters, determined by Optuna, are listed below.

Optimized Base Models (Level 0):

- **LSTM Model:** Trained on sequences of 90 timesteps. Hyperparameters: hidden_size=64, num_layers=2, dropout=0.121, lr=0.0038, batch_size=64.
- **TabNet Model:** Trained on engineered temporal features derived from the 90-timestep sequences. Hyperparameters: n_d=64, n_a=64, n_steps=8, gamma=1.05, lambda_sparse=1.78e-6, lr=0.0157, batch_size=512, cat_emb_dim=1.

Benchmark and Ablation Models: To validate our ensemble, several models representing common or powerful alternative approaches were also optimized and trained:

- **TCN Model:** Trained on sequences of 90 timesteps. Hyperparameters: num_channels=[64, 128, 192] (base=64, levels=3), kernel_size=3, dropout=0.283, lr=0.0040, batch_size=128.
- **TabNet-LSTM (Serial Hybrid):** Trained on sequences of 90 timesteps. Hyperparameters: TabNet (n_d=64, n_a=32), LSTM (hidden=64, layers=3), dropout=0.170, lr=0.00013, batch_size=32.
- **RandomForest (Benchmark):** Trained on flattened sequence data ($N \times (90 \times \text{features})$). Hyperparameters: n_estimators=500, max_depth=15, min_samples_split=5, min_samples_leaf=2.

F. Model Evaluation Metrics

The performance of all forecasting models was rigorously evaluated on the unseen test set using a suite of widely

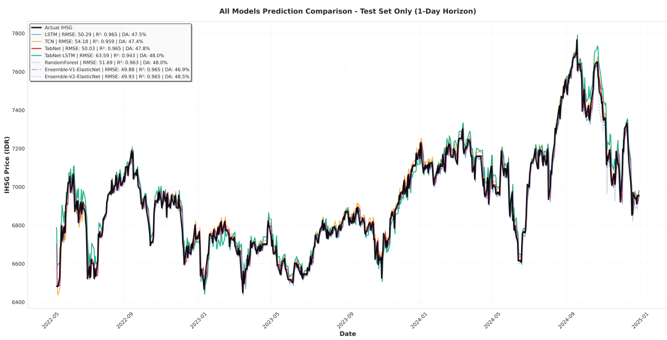

Figure 2. Model Performance Evaluation (Full Test Set).

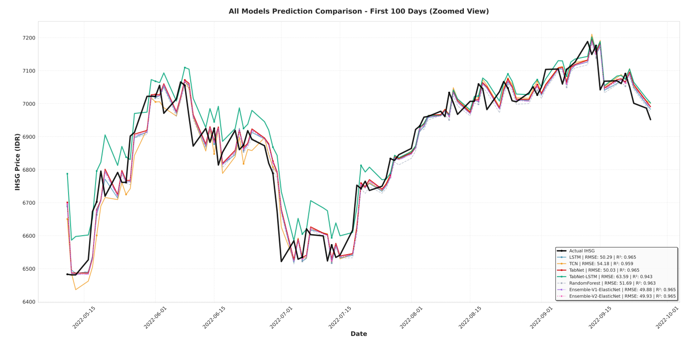

Figure 3. Model Performance Evaluation (Zoomed 100 Days).

accepted metrics. The selection of these metrics ensures a comprehensive assessment of both prediction accuracy and practical utility for financial applications.

- **Root Mean Squared Error (RMSE):** RMSE was chosen because it penalizes large errors more heavily than small ones. In finance, large prediction errors can lead to significant financial losses, making RMSE a critical indicator of model robustness against high-volatility events.
- **Mean Absolute Error (MAE):** Unlike RMSE, MAE treats all errors equally, providing a more direct and intuitive interpretation of the average error magnitude in the same units as the target variable (returns). It is less sensitive to outliers and complements RMSE by showing the model's average accuracy.
- **R-Squared (R^2):** The inclusion of R-Squared (R2) quantifies the variance explained by the model regarding financial returns. High R2 values indicate that the model captures essential market trends rather than merely reflecting a static mean response, enhancing the reliability of forecasting results [27], [28]. R-Squared is widely regarded as critical in regression analysis, providing essential insights into the model's explanatory power [27], [29].
- **Symmetric Mean Absolute Percentage Error (SMAPE):** SMAPE is useful for comparing model performance across different assets or time periods due to its scale-independent nature. SMAPE avoids the challenge of "infinite error" associated with conventional Mean Absolute Percentage Error (MAPE) when actual values are zero, ensuring a more reliable measure of performance [27], [30]. Consequently, employing SMAPE allows for a more nuanced evaluation of model efficiency, especially when forecasting financial instruments with varying value scales [30].
- **Paired t-test:** This test is used to compare the distributions of the prediction errors (specifically, the squared errors) between two models. It tests the null hypothesis that the difference in squared errors has a mean of zero, providing insight into whether one model's performance is statistically significantly different from another's.
- **Wilcoxon Signed-Rank Test:** As a non-parametric alternative to the t-test, the Wilcoxon test is used to compare the paired error distributions without assuming they follow a normal distribution. This makes it a more robust test if the error data is skewed.
- **Diebold-Mariano (DM) Test:** To further validate comparative accuracy, statistical tests such as the Diebold-Mariano (DM) test are employed. An econometric tool, the DM test assesses whether there is a statistically significant difference between the forecasting performances of two models. A low p-value (e.g., < 0.05) from the DM test enables researchers to reject the null hypothesis, indicating that one model outperforms another [31]. Such rigorous statistical validation reinforces confidence in selecting superior predictive models based on demonstrated performance, as outlined in various forecasting analyses [31], [32].

IV. Results and Discussion

A. Model Performance Evaluation

All models were evaluated on the held-out test set (689 samples). The proposed **Ensemble-V1-ElasticNet** (Stacking LSTM + TabNet) achieved the best performance overall, slightly outperforming Ensemble-V2 (which included TCN). Both ensembles ranked #1 across key error metrics (Table II). The SMAPE of 0.53% is particularly noteworthy, indicating high precision for practical applications.

The results in Table II demonstrate the stacking ensemble's value. **1) Ensemble vs. Base Models:** The Ensemble-V1 (RMSE 49.89) outperforms its base models (TabNet 50.03, LSTM 50.29), suggesting the meta-learner successfully combined their unique, non-overlapping predictive strengths. **2) Ensemble vs. Serial Hybrid:** The stacking ensemble (RMSE 49.89) is significantly better than the serial `TabNet-LSTM` (RMSE 63.59). This poor performance confirms our hypothesis that serial integration creates an information bottleneck, where the initial TabNet filter removes or over-abstracts raw temporal information essential for the LSTM's sequence modeling. **3) TCN Inclusion:** Adding the TCN (Ensemble-V2) slightly degraded performance, likely adding redundant information.

Table II
PREDICTION RESULT COMPARISON ON TEST SET

Model	RMSE	MAE	R2	SMAPE (%)
Ens-V1 (LSTM+TabNet)	**49.89**	**36.90**	**0.965**	**0.53**
Ens-V2 (All)	49.93	36.95	0.965	0.53
TabNet	50.03	37.13	0.965	0.54
LSTM	50.29	37.89	0.965	0.55
RandomForest	51.69	38.68	0.963	0.56
TCN	54.18	41.39	0.959	0.60
TabNet-LSTM (Serial)	63.59	48.50	0.943	0.70

B. Statistical Significance

To validate these findings, we conducted DM, paired t-tests, and Wilcoxon signed-rank tests. **Ensemble-V1 vs. TabNet:** The improvement is *not* statistically significant (DM p-value=0.311; Wilcoxon p-value=0.298), suggesting the engineered features for TabNet were highly effective alone. However, the ensemble still provided the lowest absolute error. **Ensemble-V1 vs. TabNet-LSTM:** The improvement is statistically significant (DM p-value < 0.001; Wilcoxon p-value < 0.001). **TabNet vs. LSTM:** The difference is *not* significant (DM p-value = 0.549; Wilcoxon p-value = 0.512), confirming they are good candidates for ensembling. These results confirm that the stacking ensemble significantly outperforms the serial hybrid.

C. Feature Importance

Feature importance from the standalone TabNet (a base learner in Ensemble-V1) provides insights into predictive drivers [33]. As shown in Fig. 4, the model prioritizes **recent lagged returns of global assets and volatility measures**. The **'S&P 500_return_lag1'** (previous day's S&P 500 return) is the most influential feature, highlighting immediate US market contagion. **'Gold_return_lag5'** (five days prior) and **'VIX_return_lag1'** (previous day's VIX change) follow. The gold lag suggests its 'safe haven' trend manifests over a week. These top features show the model relies heavily on recent global sentiment and immediate volatility over longer-term macroeconomic indicators for short-term forecasting.

D. Discussion of Limitations

Several limitations exist. First, the model relies solely on quantitative data, ignoring exogenous shocks. Financial markets are highly sensitive to qualitative information such as breaking news, political announcements, and shifts in investor sentiment. These factors are not captured here and can cause sudden, high-impact movements. Second, the daily frequency data cannot capture intraday dynamics relevant to high-frequency trading. Finally, the stacking ensemble, particularly with extensive Optuna tuning, is computationally intensive to train and deploy compared to simpler econometric models.

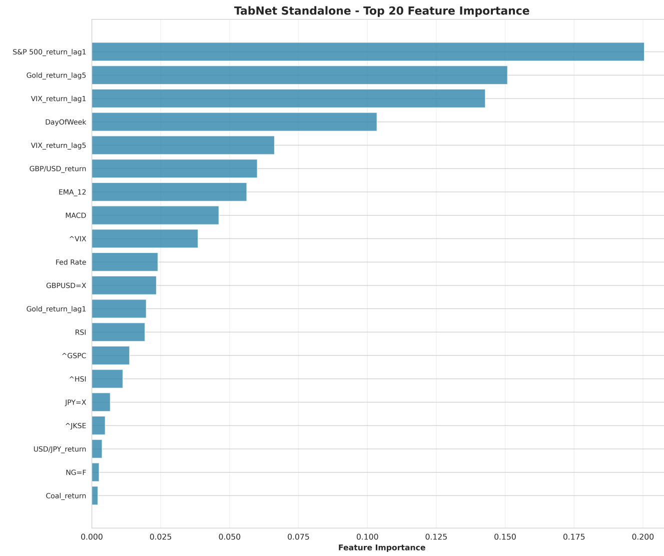

Figure 4. Top 15 Overall Most Important Features.

V. CONCLUSION

We proposed a hybrid LSTM-TabNet ensemble using an ElasticNet meta-learner to forecast the IHSG. The framework achieved a state-of-the-art RMSE of 49.89 and SMAPE of 0.53% on the test set, outperforming base models and significantly surpassing a serial TabNet-LSTM architecture. While the performance gain over the standalone TabNet was statistically marginal, the ensemble consistently provided the best point estimate. Interpretability analysis confirmed the model's reliance on recent lagged global returns ('S&P 500_return_lag1') and volatility ('VIX_return_lag1'), offering a robust, explainable forecasting tool. Future work could validate this framework on other indices or explore nonlinear meta-learners (e.g., XGBoost). Furthermore, integrating NLP-based sentiment analysis is a valuable next step. However, this process introduces significant challenges, such as the high-velocity, high-noise nature of financial news, the difficulty of accurately quantifying sentiment, and the complexity of aligning unstructured textual data with structured time-series data.

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